Project Title

Design and Simulation of an Automated Single-Axis Solar Tracking System

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1. Objective

To design a single-axis solar tracking system using SolidWorks, simulate the sun's movement using Python, and analyze the motor response using MATLAB Simulink.

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2. Software Used

Software	Purpose
SolidWorks	Mechanical design of the solar tracker
Python (Google Colab)	Sun-angle calculation and visualization
MATLAB Simulink	Motor and control system simulation

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3. Components Used

- Solar panel
- Base frame
- Rotating shaft
- DC motor (simulated)
- MATLAB Function block
- Gain block
- Transfer Function block
- Scope block

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4. Methodology

Step 1

Designed the mechanical structure of a single-axis solar tracker in SolidWorks.

Step 2

Calculated the panel angle throughout the day using Python:

Angle=(Time−6)×15\text{Angle}=(\text{Time}-6)\times15Angle=(Time−6)×15

where:

- 6 AM = 0°
- 12 PM = 90°
- 6 PM = 180°

Step 3

Plotted:

- Panel Angle vs Time
- Relative Sun Height vs Time

Step 4

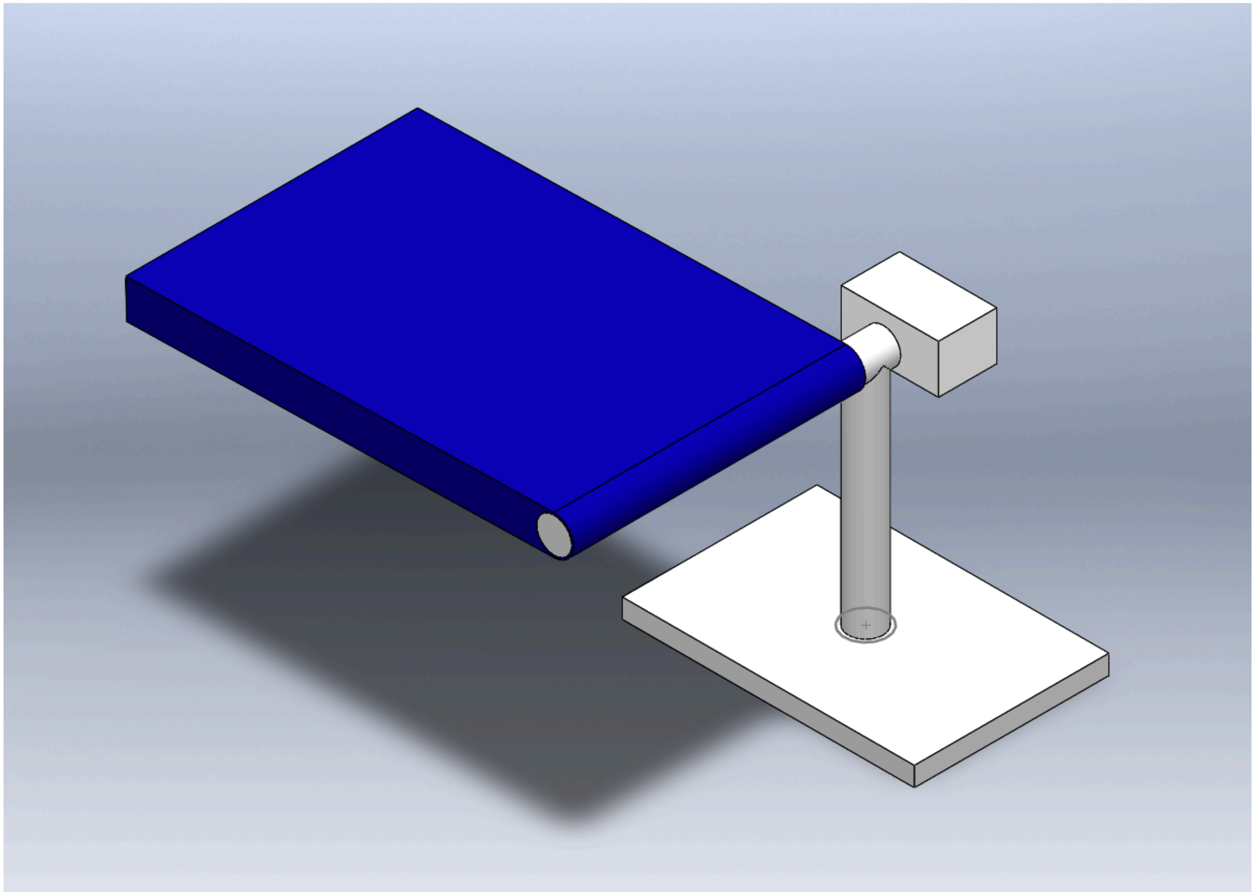
Built a Simulink model:

Clock
↓
MATLAB Function
↓
Gain
↓
Transfer Function
↓
Scope

Step 5

Observed the motor response using the Scope.

5.Solidworks Design



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6. Python Logic

Time array:

time = np.arange(6,18.5,0.5)

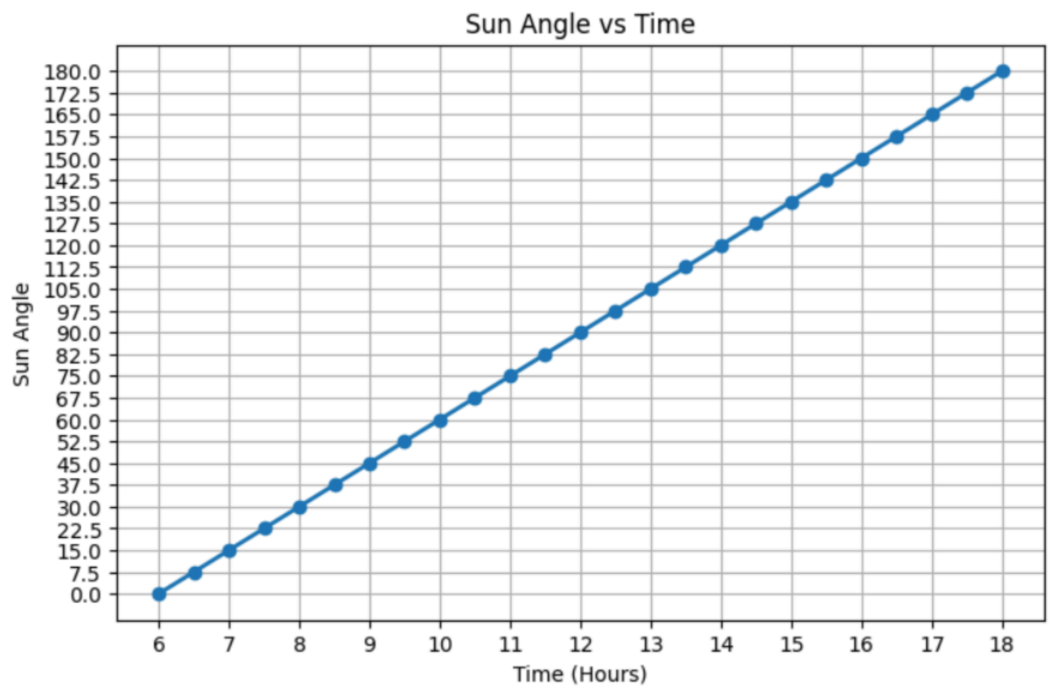
1.Panel angle:

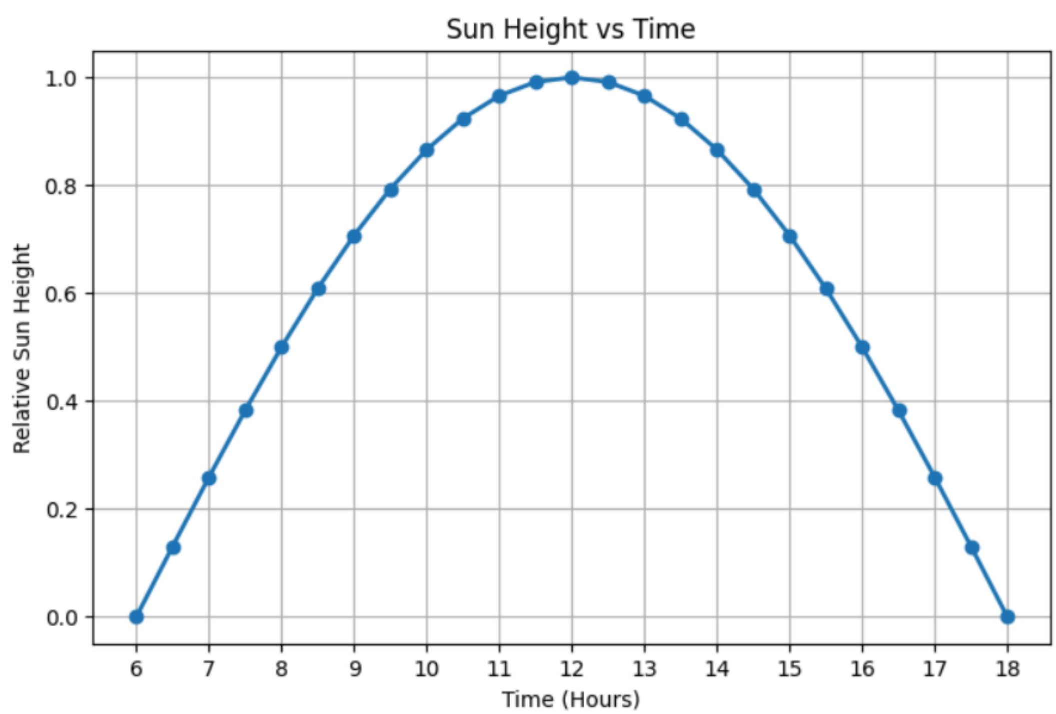
Angle=(Time−6)×15

2.Relative sun height:

Sun Height=sin(θ)

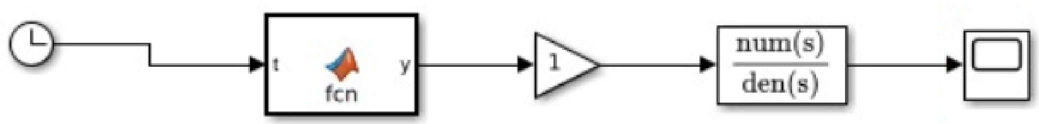
Python Output Graphs





7. Simulink Logic

- **Clock** → Generates simulation time.
- **MATLAB Function** → Converts time into desired panel angle.
- **Gain** → Represents controller/motor gain.
- **Transfer Function** → Models the dynamic response of the DC motor.
- **Scope** → Displays motor response over time.



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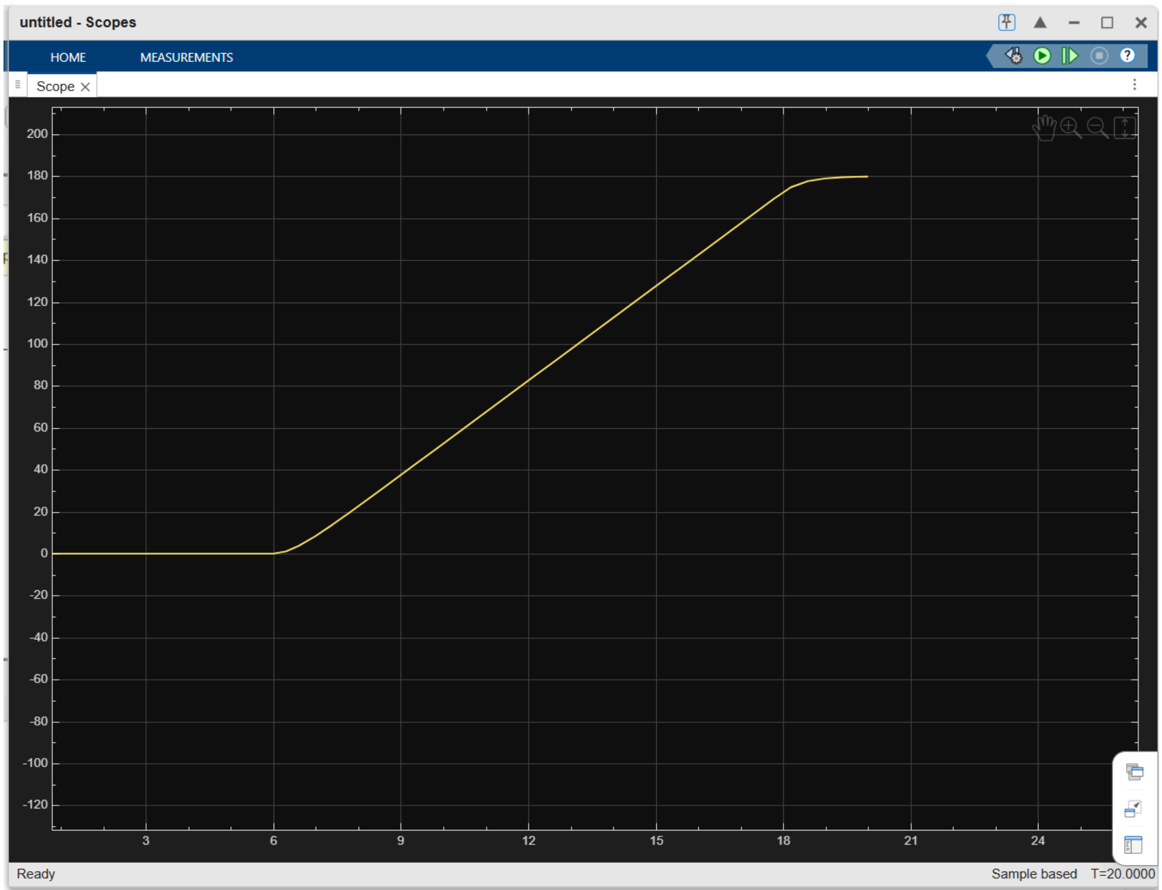
8. Transfer Function

Used:

$G(s)=1/(0.5s+1)$
It represents a first-order DC motor model.

The transfer function makes the motor rotate **smoothly** rather than changing angle instantaneously.

9.Scope Output



9. Results

- Successfully designed the solar tracker assembly.
- Generated panel-angle and sun-height graphs.
- Simulated motor response in Simulink.
- Verified smooth tracking behavior of the solar panel.

10. Conclusion

A single-axis solar tracking system was successfully designed and simulated. The integration of SolidWorks, Python, and MATLAB Simulink demonstrated the mechanical design, angle calculation, and motor dynamics required for automatic solar tracking.